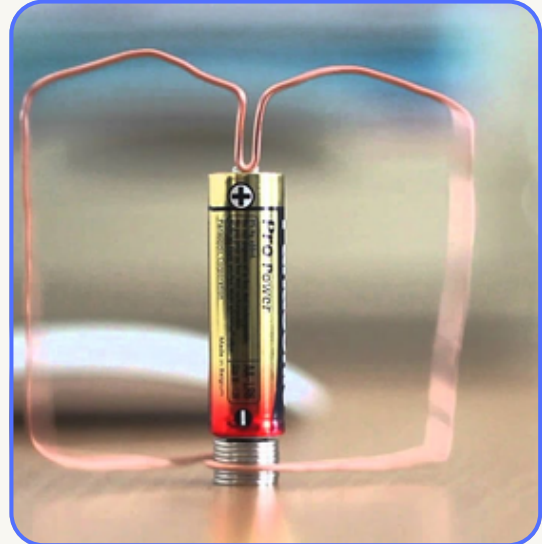


Workshop 3

HOMOPOLAR MOTOR AND DRIVE TRAIN

EQUIPMENT LIST

- 20 Alkaline batteries (AA)
- 30 Strong neodymium magnets (N52 grade)
- 1 spool of bare copper wire (1.3 mm thick)
- 1 pair of wire cutters and 1 pair of flat-nose pliers
- Templates (caps, calibration tubes)



INSTALLATION STEPS

- Cut 30-cm lengths of copper wire.
- Bend the wire to form symmetrical shapes (heart, spiral, star) with a central point.
- Place 2 to 3 neodymium magnets on the negative terminal of the battery.
- Place the wire on the battery so that the tip touches the positive terminal and the loop surrounds the magnets.
- For the train : wrap the wire around a tube to form a long spiral tunnel (solenoid).

The Autonomous Train (Wow Effect)

Attach neodymium magnets to both ends of an AA battery (with the same poles of the magnets facing each other). Slide this assembly inside the copper tube : the battery will race through the entire tube on its own, just like a magnetic train !